Comparing Consensus ADMM and Penalty Methods for Distributed Multi-Agent Trajectory Planning

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1 Introduction

Distributed optimization is central to coordinating cyber-physical systems (CPS), where agents must cooperatively solve a global objective through local computation and neighbor communication [7]. A canonical example is multi-drone trajectory planning: each drone must navigate to its own goal while avoiding collisions with nearby agents, without a central coordinator that has access to all agents' states simultaneously.

Two main classes of distributed methods handle the inter-agent collision constraint differently. The *penalty method* incorporates constraint violations directly into each agent's local objective via a quadratic penalty term, requiring only trajectory broadcast among neighboring agents. It is relatively simple to implement, but its effectiveness hinges entirely on the choice of a scalar penalty weight w: if w is too small, collisions may occur; if too large, the resulting per-agent QP becomes ill-conditioned and can even lead to *increased* constraint violations. The *Alternating Direction Method of Multipliers (ADMM)* augments the penalty with dual variables that iteratively enforce feasibility [1]. The *Consensus* variant of ADMM, designed for graph-structured multi-agent systems [2], maintains pairwise dual variables λ_{ij} that accumulate consensus errors and enforce the separation constraint asymptotically, independent of any single scalar knob.

This report compares penalty and Consensus ADMM on multi-drone trajectory planning with collision avoidance, using a centralized joint QP as the optimal reference. For each method we derive predictions about its expected parameter sensitivity from first principles and validate them against parameter sweeps in simulation. Three questions guide the investigation: (1) Do both methods converge to the centralized optimum? (2) How sensitive is each method to its tuning parameter (w for penalty, ρ for ADMM)? (3) How does each method scale with respect to the number of drones N?

2 Background and Motivation

Model Predictive Control (MPC). At each time step t, MPC solves a finite-horizon optimal control problem from the current state, applies only the first control action, advances the state, and repeats. This receding-horizon structure tolerates linearization errors in the far horizon because they are corrected at the next re-solve. For multi-agent settings, MPC naturally decomposes into per-agent subproblems coupled only through the collision constraints.

Double-integrator dynamics. Each drone is a 2D point mass with state $x_i = [p_i^\top, v_i^\top]^\top \in \mathbb{R}^4$ and control $u_i \in \mathbb{R}^2$ (acceleration). Zero-order-hold discretization at Δt gives $A = \left[\begin{smallmatrix} I & \Delta t I \\ 0 & I \end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix} \frac{1}{2}\Delta t^2 I \\ \Delta t I \end{smallmatrix}\right]$. This model is standard in the DMPC literature [4] and is justified by quadrotor differential flatness [6]: a smooth position trajectory uniquely determines the attitude and motor commands needed to track it, so the planner only needs to reason about position.

Linearized collision avoidance. The safety constraint $\|p_i(k) - p_j(k)\|_2 \geq d_{\min}$ is nonconvex, so we handle it via *sequential convex programming* (SCP). At the current estimate $\bar{p}_i(k)$, $\bar{p}_j(k)$, we form the unit normal

$$n_{ij}(k) = \frac{\bar{p}_i(k) - \bar{p}_j(k)}{\|\bar{p}_i(k) - \bar{p}_j(k)\|_2}$$

and impose the linearized half-space $n_{ij}(k)^\top (p_i(k) - p_j(k)) \geq d_{\min}$. Re-linearizing once per MPC time step keeps each agent's subproblem a convex QP.

3 Problem Formulation

We consider N drones communicating over a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \ldots, N\}$ indexes the drones and $(i, j) \in \mathcal{E}$ iff drones i and j exchange planned trajectories. We write $\mathcal{N}_i = \{j : (i, j) \in \mathcal{E}\}$ for the neighbors of drone i; collision constraints are enforced only between neighbors, since i only has access to the planned trajectory of j when $j \in \mathcal{N}_i$. Each drone i solves the finite-horizon tracking problem

$$\min_{\{x_i,\, u_i\}} \sum_{k=0}^{H} \|x_i(k) - x_i^{\text{ref}}\|^2 + \sum_{k=0}^{H-1} \|u_i(k)\|^2 \tag{1}$$

subject to dynamics $x_i(k+1) = Ax_i(k) + Bu_i(k)$, initial condition $x_i(0) = x_i^{\text{measured}}$, control bounds $\|u_i(k)\|_\infty \leq u_{\max}$ (the quadrotor's acceleration limit), and the linearized collision half-spaces for all $j \in \mathcal{N}_i$ and $k = 0, \ldots, H$. The reference x_i^{ref} is the goal state (target position, zero velocity), so the cost rewards reaching the goal and staying there.

The three methods studied are differentiated by how they enforce the coupling constraint.

4 Methods

4.1 Centralized QP (Reference)

The centralized reference combines all N per-agent problems into a single joint QP:

$$\min_{\{x_i, u_i\}_{i=1}^N} \sum_{i=1}^{N} \left(\sum_{k=0}^{H} \|x_i(k) - x_i^{\text{ref}}\|^2 + \sum_{k=0}^{H-1} \|u_i(k)\|^2 \right) \tag{2}$$

subject to dynamics, bounds, and the linearized half-spaces for *all* pairs $(i, j) \in \mathcal{E}$. The decision variable count scales as $\mathcal{O}(NH)$ and the number of coupling constraints as $\mathcal{O}(|\mathcal{E}|H)$, so the joint QP grows quadratically with N on a dense graph. We solve it with CVXPY/OSQP [8], wrapping it in a 5-iteration SCP loop that re-linearizes the collision normals.

4.2 Distributed Penalty Method

Each drone adds a quadratic penalty for collision violations to its own objective and iterates the following two steps until convergence:

1. **Primal update.** Drone i solves

$$\min_{x_i, u_i} \sum_{k=0}^{H} \big(\|x_i(k) - x_i^{\text{ref}}\|^2 + \|u_i(k)\|^2\big) + w \sum_{j \in \mathcal{N}_i} \sum_{k=0}^{H} \max\big(0,\ d_{\min} - n_{ij}(k)^\top (p_i(k) - p_j(k))\big)^2$$

   using the latest broadcast trajectories of its neighbors as fixed parameters.

2. **Broadcast.** Drone i transmits its updated planned trajectory $\{x_i^{k+1}\}$ to all $j \in \mathcal{N}_i$.

No dual variables are maintained; the only knob is the penalty weight w.

4.3 Consensus ADMM

Consensus reformulation. We lift each drone's decision variable to an *augmented* local variable $\Theta_i = (\theta_i, \{\tilde{\theta}_{ij}\}_{j\in\mathcal{N}_i})$, where θ_i is the stacked trajectory $(x_i(0), u_i(0), \ldots, x_i(H))$ and $\tilde{\theta}_{ij}$ is drone i's local copy of neighbor j's trajectory. The inter-agent coupling is replaced by the consensus constraint $\tilde{\theta}_{ij} = \theta_j$ for each edge $(i,j) \in \mathcal{E}$, and the collision half-space uses $\tilde{\theta}_{ij}$ so the per-agent QP remains purely local.

Augmented Lagrangian. Attaching a scaled dual variable λ_{ij} to each directed consensus constraint $\tilde{\theta}_{ij} = \theta_j$ gives the augmented Lagrangian [1, 2]:

$$\mathcal{L}_\rho = \sum_{i\in\mathcal{V}} g_i(\Theta_i) + \sum_{i\in\mathcal{V}} \sum_{j\in\mathcal{N}_i} \left[\lambda_{ij}^\top (\tilde{\theta}_{ij} - \theta_j) + \tfrac{\rho}{2} \left\| \tilde{\theta}_{ij} - \theta_j \right\|_2^2 \right], \tag{3}$$

where $g_i(\Theta_i)$ packages everything that is local to drone i:

$$g_i(\Theta_i) = \underbrace{\sum_{k=0}^{H} \|x_i(k) - x_i^{\text{ref}}\|^2 + \sum_{k=0}^{H-1} \|u_i(k)\|^2}_{\text{tracking + control cost}} + \mathcal{I}_{\mathcal{C}_i}(\Theta_i), \tag{4}$$

and $\mathcal{I}_{\mathcal{C}_i}$ is the $\{0, +\infty\}$ indicator of the feasible set $\mathcal{C}_i$ defined by the dynamics $x_i(k+1) = Ax_i(k) + Bu_i(k)$, the initial condition $x_i(0) = x_i^{\text{measured}}$, the control bounds $\|u_i(k)\|_\infty \leq u_{\max}$, and the linearized collision half-spaces between drone i's own position and its local belief $\tilde{\theta}_{ij}$ for every $j \in \mathcal{N}_i$.

Algorithm. The three-step iteration is:

Step 1. Primal (parallel over agents). Each drone i solves jointly for its own trajectory and its beliefs about its neighbors:

$$\Theta_i^{k+1} = \arg\min_{\Theta_i} \left(g_i(\Theta_i) + \sum_{j\in\mathcal{N}_i} \left[(\lambda_{ij}^k)^\top \tilde{\theta}_{ij} + \tfrac{\rho}{2} \left\| \tilde{\theta}_{ij} - \theta_j^k \right\|_2^2 \right] \right). \tag{5}$$

We write the components of the minimizer as $\Theta_i^{k+1} = (\theta_i^{k+1}, \{\tilde{\theta}_{ij}^{k+1}\}_{j\in\mathcal{N}_i})$.

Step 2. Communicate. Each drone broadcasts its own θ_i^{k+1} to its neighbors.

Step 3. Dual update (parallel over directed edges).

$$\lambda_{ij}^{k+1} = \lambda_{ij}^k + \rho \left(\tilde{\theta}_{ij}^{k+1} - \theta_j^{k+1} \right). \tag{6}$$

This pattern requires no central aggregation; all communication is local to the graph edges. Convergence is declared when the primal residual $r^k = \|\tilde{\theta}_{ij}^k - \theta_j^k\|$ and the dual residual $s^k = \rho\|\theta_j^k - \theta_j^{k-1}\|$ fall below tolerances ε^{pri} and $\varepsilon^{\text{dual}}$ [1].

Implementation notes. A soft slack on each collision half-space with weight $w_{\text{coll}} = 10^7$ prevents infeasibility crashes during early iterations when linearizations may be degenerate; this is a numerical safety valve and does not participate in the feasibility-driving mechanism. The per-agent QPs in ADMM and the penalty method are solved with Clarabel [3]; the centralized joint QP uses OSQP [8].

5 Predictions

Penalty: feasibility versus conditioning. The penalty method does not guarantee convergence to feasibility. As $w \to \infty$ violation scales by $\mathcal{O}(1/w)$, but the per-agent Hessian condition number grows proportionally to w, so the outer iteration's per-step progress shrinks and the loop exhausts

its iteration budget and terminates with non-negligible residual violations. This leads to a characteristic U-shaped relationship between w and constraint violation: when w is small, collisions persist because the penalty is too weak; when w is large, numerical ill-conditioning dominates and again prevents effective convergence; only an intermediate range of w yields satisfactory performance, where constraint enforcement and numerical stability are balanced.

ADMM: dual variables decouple feasibility from conditioning. Boyd et al. [1] shows that Consensus ADMM converges for any $\rho > 0$ under convex local costs (which our setting satisfies); the role of ρ is therefore not to drive feasibility but to shape the convergence rate. Concretely, the dual update (6) is an integrator: at a fixed point, $\lambda_{ij}^{k+1} = \lambda_{ij}^{k}$ forces exact consensus $\tilde{\theta}_{ij} = \theta_j$. Collision avoidance is enforced through a two-step chain: First, the hard half-space inside g_i keeps drone i safe relative to its belief $\tilde{\theta}_{ij}$ at every iteration. Second, λ_{ij} corrects the belief to match the neighbor's true broadcast state. Together they yield true inter-agent separation in the limit. The dual variable λ_{ij} thus drives the feasibility, while ρ enters only as a step-size for the dual update and a curvature term in the per-agent QP. The two jobs are handled by two separate quantities, unlike the single weight w in the penalty method. This indicates that ADMM should be much less sensitive to ρ than penalty is to w: the violation curve should stay near zero across a wide range of ρ, even though iteration count required to converge may exhibit a U-shape from rate sensitivity.

6 Experimental Setup

We evaluate on two scenarios that stress the collision-avoidance coupling at increasing scale:

S1. Four-drone ring ($N = 4$). Four drones start at 90° intervals on a circle of radius 3 m and swap to antipodal goals. All four trajectories cross near the origin. The communication graph is a ring, so each drone enforces collision constraints only with its two ring neighbors; the two diagonal pairs are unconstrained (they are further apart and less prone to collision in this geometry). Parameters: $H = 25$, $\Delta t = 0.2\,\mathrm{s}$, $u_{\max} = 3\,\mathrm{m/s^2}$, $d_{\min} = 0.5\,\mathrm{m}$. Penalty uses $w = 10^3$; ADMM uses $\rho = 5$.

S2. Eight-drone complete graph ($N = 8$). Eight drones start at 45° intervals on a circle of radius 3 m and swap antipodally. The communication graph is complete: every pair of drones enforces a collision constraint. Parameters: $H = 30$, $\Delta t = 0.2\,\mathrm{s}$, $u_{\max} = 3\,\mathrm{m/s^2}$, $d_{\min} = 0.5\,\mathrm{m}$. Penalty uses $w = 2 \times 10^3$; ADMM uses $\rho = 300$. The straight-line warm-start is degenerate at $N = 8$ (all drones pass the origin at the same step), so ADMM is seeded with the penalty output to break symmetry; the centralized solver uses the same seed.

For both scenarios we report: the number of outer iterations (primal broadcasts) to convergence, the final objective $J^\star$, the minimum pairwise distance $\min_{i<j,k} \|p_i(k) - p_j(k)\|_2$, and the constraint violation (maximum distance by which any pairwise separation falls below $d_{\min}$).

The SCP outer loop converges to a local KKT point of the nonconvex collision-avoidance problem [5], not the global optimum. Antipodal swaps have multiple local minima (clockwise vs. counterclockwise orbits). "Centralized" in this report therefore refers to the local optimum reached by SCP from the shared seed above, not the global optimum of the original problem. All three methods optimize against the same linearized collision constraints from the same seed, so the comparison isolates the effect of the solver rather than the constraint model.

We record violation and iteration count for two sensitivity sweeps on S1: penalty over $w \in [10^2, 10^4]$ and ADMM over $\rho \in [10^{-1}, 10^3]$, with eight and nine log-spaced values respectively. Both use a 10,000-iteration outer-loop cap.

7 Numerical Results

7.1 S1: Four-Drone Ring

Figure 1 shows the planned trajectories for S1. All three methods produce visually smooth paths that reach the goal positions, and the three objective values agree to within 0.04%. The critical

distinction emerges in the constraint column of Table 1: the penalty method's minimum pairwise distance is 0.483 m, which is 17 mm inside the exclusion disk. This is a real violation: it occurs at a discrete MPC step where two drones are geometrically inside each other's safety radius. ADMM and the centralized solver both respect the 0.5 m boundary to solver tolerance. In terms of iteration count, ADMM converges in 44 outer rounds versus 99 for penalty, reflecting the additional feedback from the dual variables.

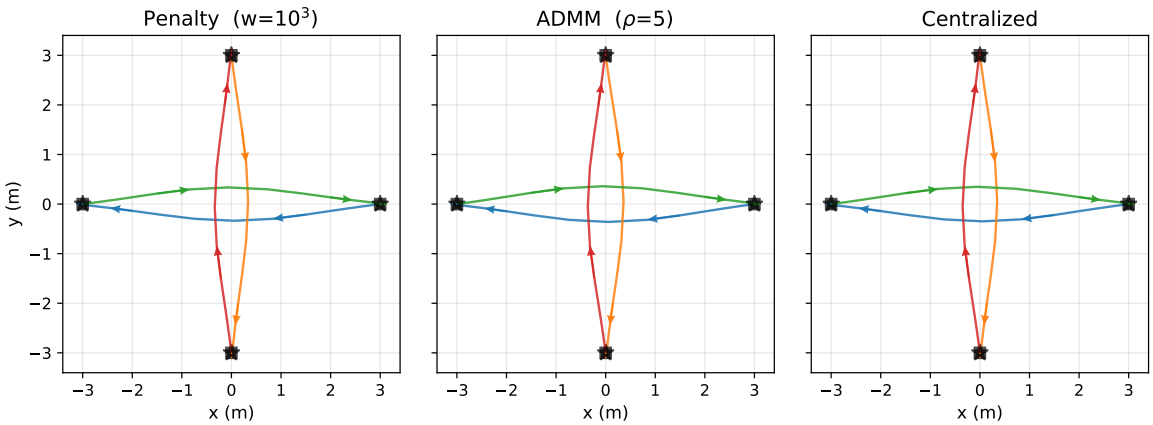

Figure 1: S1 ($N = 4$, ring graph): planned trajectories from penalty (left), Consensus ADMM (center), and centralized (right). Filled squares are start positions; hollow stars are goals. All three reach their goals, but penalty's minimum pairwise distance falls 17 mm below the safety threshold.

Table 1: S1 ($N = 4$, ring graph) results. Violation is $\max_{i<j,k} \max(0, d_{\min} - \|p_i(k) - p_j(k)\|_2)$.

Method	Iterations	$J^\star$	Min. dist. (m)	Violation (m)
Penalty ($w = 10^3$)	99	8944.8	0.483	1.67×10^{-2}
ADMM ($\rho = 5$)	44	8948.6	0.517	0
Centralized	5	8946.5	0.500	0

7.2 Penalty Sensitivity to w

Figure 2 sweeps w over more than two decades. Violation shows the predicted U-shape: at small w the collision cost is too weak to deter overlap ($\sim$0.42 m at $w = 10^2$); at large w the per-agent Hessian becomes ill-conditioned and the outer iteration stalls, leaving violation at $\sim$0.24 m for $w = 10^4$. Both failure modes are visible in the iteration panel: the iteration count climbs sharply at large w as the ill-conditioned QP slows convergence. The sweet spot is narrow, and a single decade in either direction substantially degrades feasibility.

7.3 ADMM Sensitivity to ρ

Figure 3 repeats the sweep for ADMM, sweeping ρ over four decades on the same scenario and the same y-axis. The contrast with penalty is stark: violation stays near zero across the entire range, with only a small $\sim$0.017 m floor at $\rho \leq 3$. ADMM does exhibit a U-shape in iteration count – ρ is a rate-control parameter, and convergence slows at both extremes – but the rate sensitivity does not translate into a violation penalty: across nearly four decades of ρ the algorithm achieves true feasibility, validating the prediction that the dual variable carries the feasibility-driving role separately from ρ.

7.4 S2: Eight-Drone Complete Graph

Figure 4 shows S2 trajectories. The feasibility gap between penalty and ADMM grows dramatically at $N = 8$. Penalty ($w = 2 \times 10^3$) leaves a minimum pairwise distance of 0.175 m, a violation of 0.325 m, roughly 20$\times$ larger than the S1 violation at a comparable w. ADMM with $\rho = 300$ matches

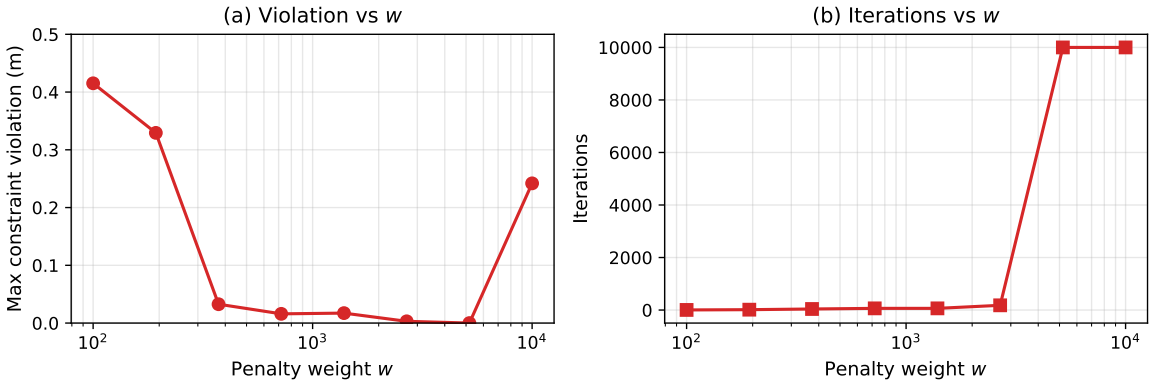

Figure 2: Penalty method sensitivity to w on S1. Left: constraint violation versus w. Right: iteration count versus w. The U-shape in violation reflects the fundamental tension between softness (left tail) and ill-conditioning (right tail).

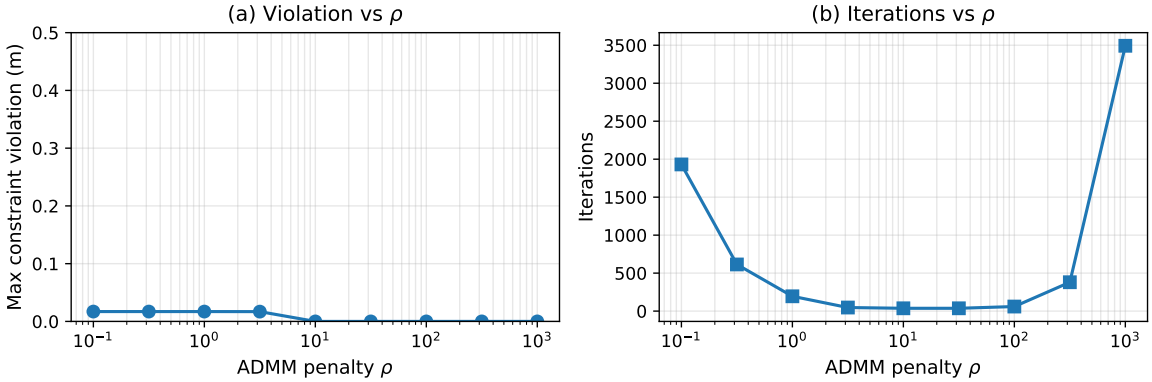

Figure 3: ADMM sensitivity to ρ on S1 (same y-axis as Figure 2). Left: violation stays near zero across four decades of ρ, with a small floor at $\rho \leq 3$. Right: iteration count is U-shaped, reflecting ρ's role as a rate parameter rather than a feasibility knob.

the centralized objective to 0.03% and achieves a minimum distance of exactly 0.500 m, with residual violation below 10^{-4} m (solver tolerance). Table 2 summarizes the numbers.

The larger ρ required at $N = 8$ (300 vs. 5 at $N = 4$) is consistent with theory: each agent has 7 neighbors in the complete graph versus 2 in the ring, so the consensus penalty term in (3) has more terms. A larger ρ is needed to maintain the same per-iteration dual feedback magnitude per constraint.

Table 2: S2 ($N = 8$, complete graph) results.

Method	Iterations	$J^\star$	Min. dist. (m)	Violation (m)
Penalty ($w = 2 \times 10^3$)	20	26981	0.175	3.25×10^{-1}
ADMM ($\rho = 300$, seeded)	111	17125	0.500	1.2×10^{-4}
Centralized	5	17120	0.500	≈ 0

Note that at S2, the penalty method's objective (26981) is also substantially higher than the centralized value (17120). This is not a coincidence: penalty trajectories are not collision-free, so they represent infeasible points of the constrained problem and do not admit a meaningful comparison of feasible-solution quality.

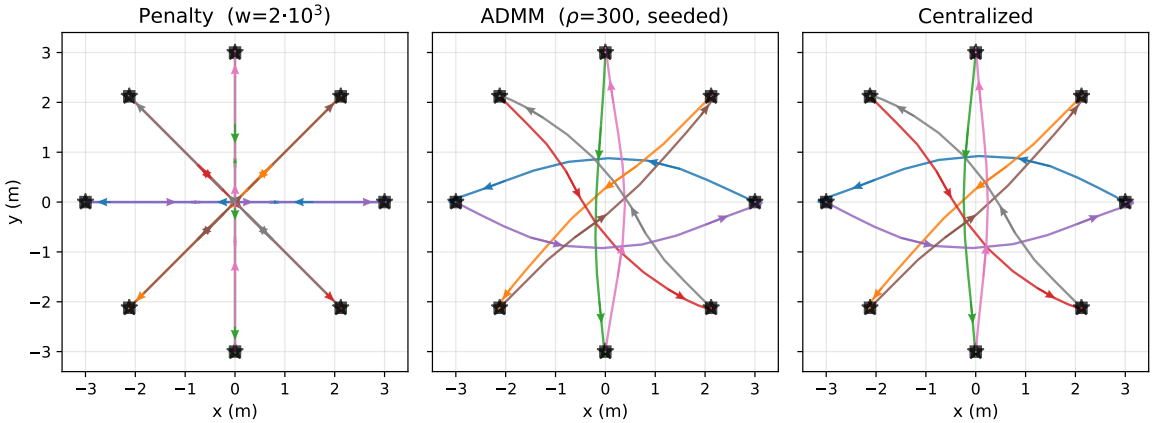

Figure 4: S2 ($N = 8$, complete graph): planned trajectories from penalty (left), Consensus ADMM (center), and centralized (right). Penalty drones overlap by up to 0.33 m; ADMM and centralized respect the safety margin.

8 Discussion

Feasibility guarantee matters. A 0.04% difference in objective value between penalty and centralized on S1 would be easy to dismiss as negligible. But the objective measures tracking plus control cost — it does not measure whether the drones physically collide. The minimum pairwise distance, which is the safety-critical metric, shows a real violation hidden by the soft cost. Objective agreement is therefore not a reliable proxy for correctness in safety-constrained planning.

The U-shape in w is structural, not incidental. The penalty method's sensitivity to w is not a tuning failure; it is a consequence of trying to use a single scalar for two conflicting roles. A small w cannot deter collisions; a large w corrupts QP conditioning. The sweet spot between them shrinks as N grows because more coupling constraints compete for the same penalty budget. This structural tension does not disappear with better initialization or tighter solver tolerances.

ADMM's dual variable is the key mechanism. The dual update $\lambda_{ij} \leftarrow \lambda_{ij} + \rho(\tilde{\theta}_{ij} - \theta_j)$ accumulates the consensus residual across iterations like an integrator. At a fixed point the increment is zero, forcing $\tilde{\theta}_{ij} = \theta_j$ and, in turn, true inter-agent separation through the hard half-space inside g_i. This decoupling of feasibility-driving (dual) from QP-conditioning (ρ) is why ADMM is far less sensitive to its tuning parameter than penalty: across nearly four decades of ρ the achievable violation stays at zero, while penalty's violation curve has peaks at both ends of w.

Scalability and limitations. Both distributed methods require N per-agent QPs of size $\mathcal{O}(H)$ per outer iteration with $\mathcal{O}(|\mathcal{E}|)$ trajectory broadcasts; the communication is purely local. The centralized joint QP has size $\mathcal{O}(NH)$ and grows quadratically with N; it serves only as a reference. Luis et al. [4] demonstrated a similar DMPC pattern on a 20-quadrotor hardware swarm, suggesting the scheme transfers to physical systems. Three limitations merit follow-up: (1) an adaptive ρ scheme [9, 10] would reduce ADMM sensitivity to the initial ρ^0; (2) per-iteration SCP re-linearization would tighten the convex approximation at higher computational cost; (3) the evaluation does not include noise, packet loss, or dynamic obstacles.

9 Conclusion

We compared Consensus ADMM and distributed penalty method for multi-drone trajectory planning with collision avoidance, using a centralized QP as the reference. Table 3 summarizes the trade-offs. On the 4-drone scenario (S1), penalty achieves near-optimal cost but violates the safety constraint by 17 mm. On the 8-drone scenario (S2), the violation grows to 325 mm, roughly 65% of the safety radius, while ADMM matches the centralized optimum to 0.03% with no practical violation. The

Table 3: Method comparison summary.

	Penalty	**ADMM**	**Centralized**
Implementation complexity	Low	Medium	Low
Per-iteration work	N small QPs	N larger QPs	1 joint QP
Feasibility guarantee	None (w-dep.)	In the limit	Hard
Tuning knobs	w	ρ	Solver only
Parallelism	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N^2)$ central

feasibility gap between the two methods grows approximately 20× going from $N = 4$ to $N = 8$.

The core finding is that ADMM's dual-variable feedback pays off exactly where the penalty method's single weight w runs out of room. That gap grows with the number of coupled agents and the density of the communication graph, making ADMM the preferred choice whenever safety constraints must be reliably satisfied in a multi-agent setting.

10 Team Contributions

Christian Yu implemented Consensus ADMM, including the augmented Lagrangian formulation, the three-step primal-dual iteration, and the convergence and sensitivity experiments sweeping ρ. Yuyang Wu developed the drone dynamics model, the MPC formulation with double-integrator discretization, the centralized QP baseline, and the trajectory visualizations. Yen-Ru Chin implemented the distributed penalty method baseline, designed the experimental scenarios S1 and S2, and led the writing of the report. All three members contributed to the literature review, the final presentation, and revisions to the report.

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